

The Model Store a certified retrieval machine

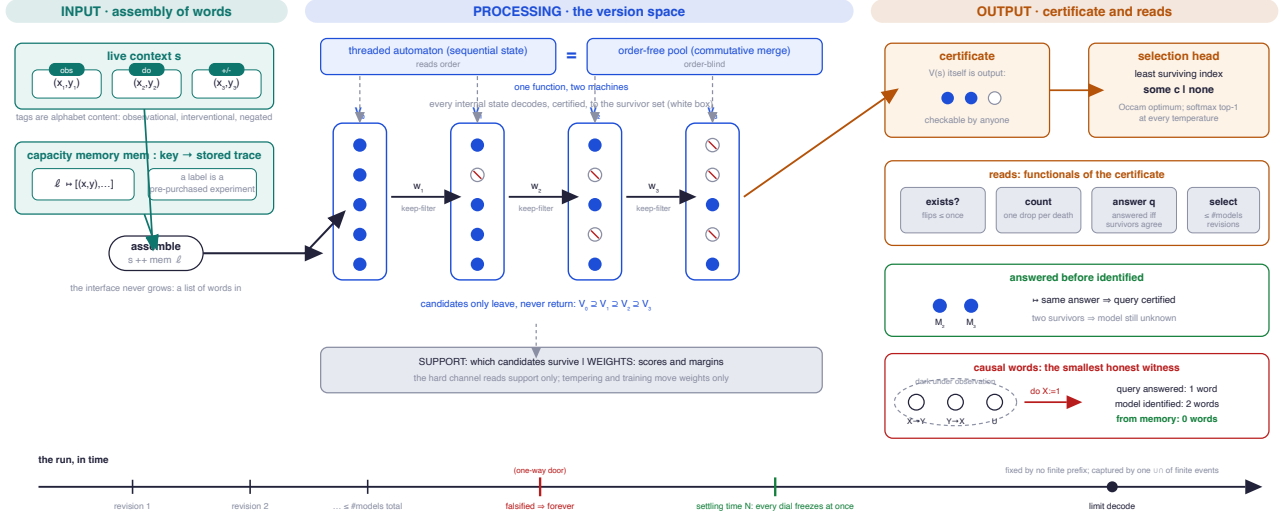
A constructed object. Every property below is a machine-checked theorem, exact rather than asymptotic; the stated instances execute inside the proof checker.

The object. A finite list of candidate models $c_1, \dots, c_k$, each a program $Dc : X \rightarrow Y$, operated as a retrieval machine over observation words.

INPUT. A context: a finite list of words (x, y) , assembled from live evidence and from memory (a label is a key whose stored trace is appended to the context). Tags on words carry observational, interventional, and negated readings. The interface is this one list type; it never grows.

PROCESSING. Maintain the *version space* $V(s)$: the candidates consistent with every word so far. One observation is one keep-filter; the space only shrinks. Computed by a sequential state machine or by an order-free commutative merge, which agree on every context. Every internal state decodes, with a checked certificate, to $V(s)$.

OUTPUT. The least-indexed survivor (or *none*), together with $V(s)$ itself as the certificate. All further answers are reads of the certificate: existence, counts, and per-model query functionals, each certified against every survivor.



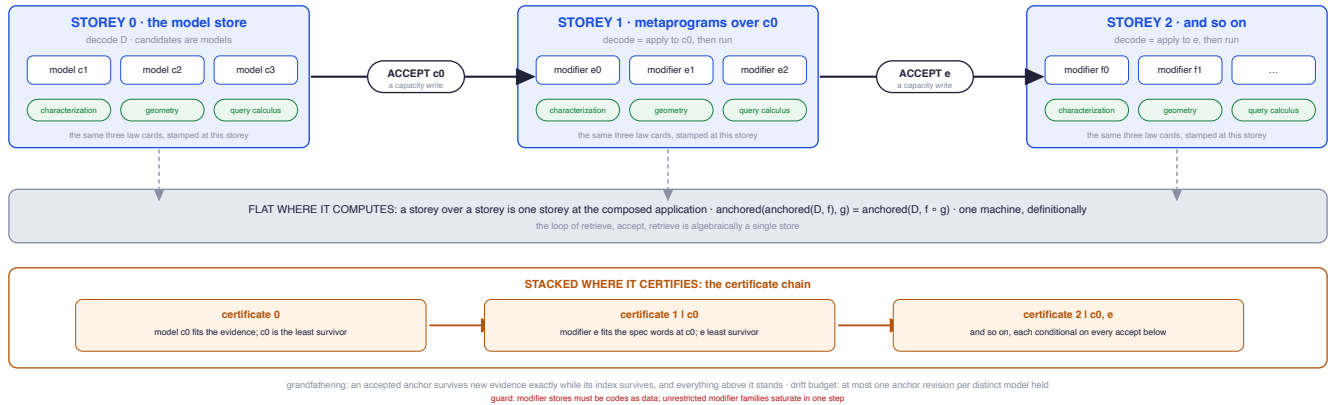
PROVEN PROPERTIES

- P1. One function, two machines.** The threaded state machine and the order-free merge compute the same answer on every context, for every store, with no side condition.
- P2. White box.** Every reachable internal state decodes to the survivor set, and any two faithful decodings agree along the run: there is no hidden internal freedom.
- P3. Exact capacity price.** The smallest machine reproducing the store's answers has size exactly the number of reachable survivor configurations. Sequential state count and parallel width times precision are the same price in two currencies, exchanged by complementation.
- P4. Order is a resource.** Order-free processing recovers everything determined by the survivor set and provably cannot recover the order of eliminations; a sequential pass recovers it exactly.
- P5. Exact revision budget.** The answer changes at most once per distinct model held, and an adversarial data order spends the whole budget. Duplicate encodings of one model never cost a revision.
- P6. Universal revision inheritance.** The survivor set itself changes at most k times, and every read of it inherits that budget. The existence read flips at most once: refutation is permanent, consistency forever provisional.
- P7. Temperature-free Occam selection.** The output is at once the least-indexed survivor, the optimum under every complexity-decreasing weighting, and the softmax top choice at every temperature. Hardening the soft regime moves only the weights, never the answer.
- P8. Dark knowledge located.** The weight stratum (scores, margins) is not a function of the survivor trajectory: two runs with identical survivor sets at every step can carry different weights. Everything the soft channel exposes beyond the answer lives in the losers.
- P9. Simultaneous settling.** On any unbounded stream the survivor set is eventually constant, and every readable dial freezes at one common time.
- P10. Identification at the limit.** If some candidate fits the whole stream, the limit answer is a candidate consistent with every prefix, reached within the revision budget. A target outside the list is approximated by the simplest consistent candidate, never identified: the machine's one boundary.
- P11. The limit, exactly bracketed.** No finite context determines the limit answer, yet the limit event is fully captured by one countable union of countable intersections of finite-context events.
- P12. Answered before identified.** A query is answered the moment all survivors agree on its value, with a certificate, possibly long before the model is pinned; answering can be strictly cheaper than identifying.
- P13. Labels are pre-purchased experiments.** A label's whole effect is the intersection law of its stored trace. It cannot separate candidates its stored words cannot kill, and a query can cost zero live words when the right experiment is already in memory.
- P14. Evidence of every polarity is alphabet content.** Demands to fail on a word, and words read under intervention, enter by recoding the alphabet; every property above transfers to the recoded store unchanged. On the smallest causal store, mechanisms indistinguishable under observation are separated by one interventional word: the query is answered at price one, the model identified at price two, and the same query costs zero from memory.

Kernel-checked in Lean 4 over Mathlib; every theorem's axiom footprint is the standard triple or less; the causal and exchange instances above are computed end-to-end by the checker's decision procedure. Selection, prices, and budgets are equalities. The weight stratum, distributional refinements, and the exact price law of experiments are stated obligations of the next construction, not properties claimed here.

The Second Storey metaprogram retrieval, and the limits of finiteness

The argument. A retrieved model can be accepted as the *anchor* of a next store, whose candidates are modifiers: codes that apply to the anchor and run. The anchored decode is a decode, so a store of modifiers is a store, and retrieval one storey up is the same machine. The query “here is my model, retrieve the modifier meeting my new conditions without touching the structure I accepted” is ordinary retrieval at the second storey; the loop of retrieve, accept, retrieve closes into a cascade.



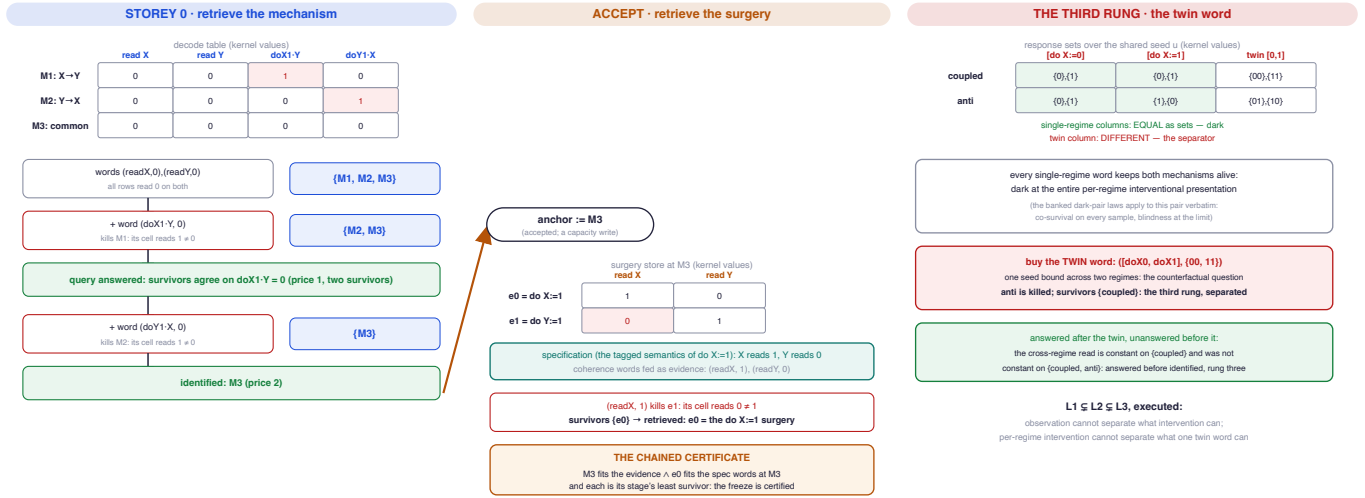
- M1. Laws at every storey.** Every anchored store, for every anchor and every family of modifiers, carries the complete characterization, the full geometry, and the whole query calculus of pages one and three. The analysis of the machine is storey independent: one theorem covers all levels at once.
- M2. Flat where it computes, stacked where it certifies.** Composing anchorings is, definitionally, one anchoring at the composed application: the cascade is a single store at the level of computation. What stacks is the certificate chain: each acceptance freezes an anchor, and every downstream certificate is conditional on the accepts below it.
- M3. The chained certificate carries the freeze, at every depth.** A cascade of any length is certified at every storey: each stage’s modifier fits its evidence at its anchor and is that stage’s least survivor, each certificate conditional on the accepts below it. Running one cascade after another is running their concatenation; the two-stage certificate is the single-storey reading. The greedy selection is part of the certificate, not an unstated convention.
- M4. Grandfathering, with an exact drift budget.** An accepted anchor survives new evidence exactly while its index survives, and while it does, everything built on it stands unchanged. The anchor itself is revised at most once per distinct model held: the model-grain revision law, read at the base store.
- M5. Correctness is evidence.** A modifier meets a specified behavior on given inputs exactly when it fits the words assembled from that specification. Checking a candidate surgery against its specification is a run of the machine on specification words, and the resulting certificate says so: agreement with the specification, never truth about the world.
- M6. Where finiteness ends.** At a countably infinite store, identification on realizable streams survives (the ordering of the indices replaces the finite count), and the per-word laws survive with it. What dies is exactly the uniform part: no store-size bound on revisions exists, unconditional settling fails off realizability, and selection is no longer computable, only specifiable. Finiteness is the price of the uniform, exact laws; it was never the price of identification.

Modifier stores obey the machine’s own guard: candidates are codes as data (unrestricted modifier families collapse the storeys in one step; grammar-bound modifiers climb, page four). Retrieval over level-sorted modifiers returns the least-level fit; anchoring adds nothing to the selection price, and execution pays a per-storey capacity product at every depth, with the measurability side discharged on standard domains. Adjacent frameworks (adapter fine-tuning, library learning, partial evaluation, knowledge compilation, specification monads) share the shape; none certifies its retrieval or prices it exactly.

The Witness, by Construction

one causal question through every storey

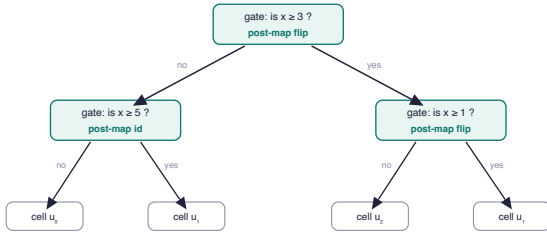
Every table cell and every survivor-set transition in the figure below is evaluated inside the proof checker. The figure does not illustrate the argument: it exhibits it. Reading the tables and checking each elimination by hand reproduces the proofs. Three rival mechanisms are indistinguishable under observation; one correctly chosen intervention answers the pressing question while two suspects still stand; the second intervention identifies the mechanism; the accepted mechanism anchors a store of surgeries from which the requested modifier is retrieved on specification words, with a joint, conditional certificate; and two mechanisms agreeing under every separate intervention are told apart only by the twin question that binds one hidden seed across two regimes.



The Witness the router forest, as built

The realization. The machine of page one is abstract: any candidate list, any decode. The witness is the concrete object it was discovered inside: a routed network read as a *forest of program trees*. A candidate is a finite tree; a gate at each node chooses a child by a threshold on the input; an elementary compute cell sits at every base and a finite post-map at every node, so the network's compute family is welded into the grammar rather than bolted on. The machine's retrieval runs over this forest verbatim, and everything on this page is a theorem about the drawn objects.¹

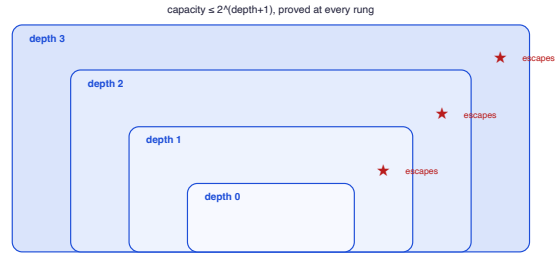
A PROGRAM IS A TREE · drawn from the construction



one candidate – one finite tree: a gate picks a child by a threshold on the input, the chosen child answers, the node's post-map finishes. The compute family is constitutive: an elementary cell at every base, a finite post-map at every node.

free selection is forbidden by the machine's own theorem: with unrestricted selectors, one step reaches every function, so candidates must be trees (data).

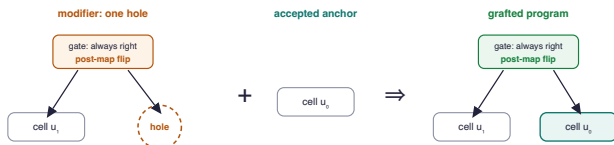
THE TOWER · strict at every rung, priced at every rung



every rung strictly larger, on the counting world and on the real-vector world; the union over all depths never exhausts the function space. Retrieval over depth-sorted candidates returns a least-depth fit: built-in simplicity bias.

the machine of page one runs verbatim over this forest: same retrieval, same laws.

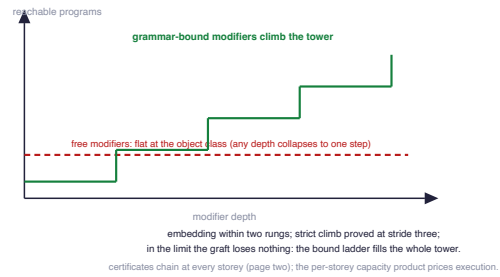
MODIFY BY GRAFTING · the first rung climbs, in one picture



grafting is part of the grammar: a plugged modifier is a tree, so modified programs inherit every law. Here the gate always routes to the graft and the post-map flips it: the grafted program disagrees with its anchor on every input. One modifier, one rung.

retrieval over modifiers = grafting: the accepted anchor enters every candidate by the hole.

FREE IS FLAT, BOUND CLIMBS · the dichotomy



- W1. Trees are forced, not chosen.** If selection over the compute family were a free function of the input, one selection step would already realize every function: no hierarchy could exist. Restricting selection to gates over tree-shaped candidates is what makes depth mean anything.
- W2. The tower is strict at every rung, twice.** Depth- $\ell+1$ trees realize strictly more than depth- ℓ trees, at every ℓ , on both worlds; one reflection step over any rung escapes that rung; and the union over all depths still never exhausts the function space.
- W3. Capacity is graded by depth.** The depth- ℓ class shatters no set of size $2^{\ell+1}+1$: the price of a rung is known before it is bought, and retrieval over depth-sorted candidates returns a least-depth fit, so the machine's selection is a simplicity bias with a certified price tag.
- W4. Modifiers live in the same grammar.** A modifier is a tree

- with one hole; applying it to an accepted anchor is grafting; a grafted modifier is again a tree. Retrieval at an anchored store of modifiers *is* grafting, and each modifier frame carries its own post-map and cells: the compute joint is present at every storey, not only the first.
- W5. The graft is cheap and loses nothing.** Everything at depth ℓ reappears among grafted modifiers within two rungs; the grafted ladder provably climbs (strictly within every three rungs, and already at its first rung); and in the limit the grafted ladder reaches exactly the whole tower.
- W6. The dichotomy.** Free modifier families are flat: any depth of anchoring reaches exactly the object class, because composed anchorings collapse to one. Grammar-bound modifiers climb the tower. Depth is a resource exactly when modifiers are data in the object's own syntax.

Stated, not claimed: whether the two-rung graft embedding and the three-rung strict climb are sharp (the drawn one-rung climb is proved at the first rung), and whether the threshold gates read literally as scored routers, which would let the per-storey price law of page two apply to the tower word for word. Every drawn tree, gate, and graft above is a term in the verified construction.

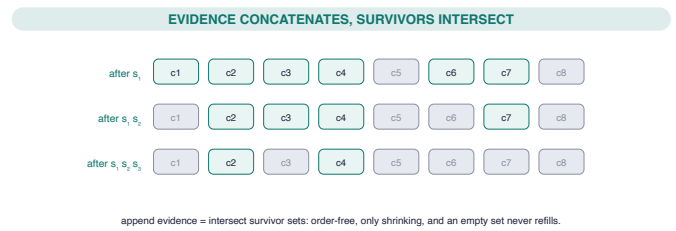
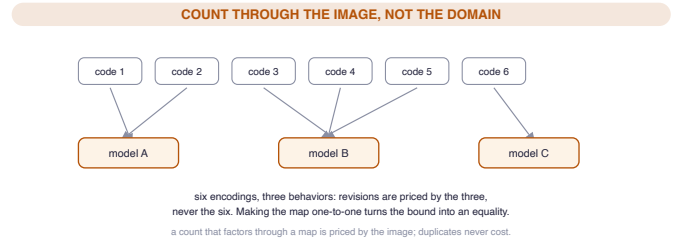
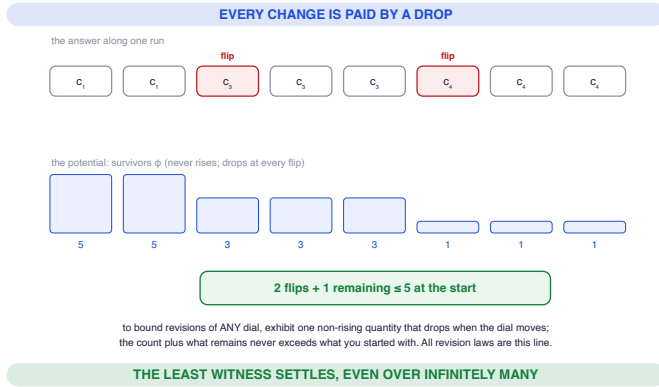
¹Capacity grading is a Sauer-Shelah-type bound per rung; the strict hierarchy is proved on the counting world and on the real-vector world with affine threshold gates. The saturation theorem that forces tree-shaped candidates is a Cantor-style diagonal argument at the selection layer.

Methods, I ways of counting

The machine's laws are instances of a small number of free-standing ideas. Each is stated here as its own tool, usable on any system, with the machine as one worked instance; each is a verified general theorem, with its instances derived from it rather than proved again.

Every change is paid by a drop. Watch any dial of a running process. If some quantity never rises and strictly falls each time the dial moves, then the number of moves plus what remains of the quantity never exceeds what it started at. Bounding revisions of anything is exhibiting such a quantity; for the machine, the dial is the answer and the quantity is the survivor count, giving the exact revision budget and its inheritance by every read.²

Count through the image, not the domain. When a count factors through a mapping, the price is set by the image, never the domain: duplicates are free. If the mapping is made one-to-one the bound becomes an equality. For the machine this is why revisions are priced by distinct behaviors rather than by encodings, and why the price is exact once duplicate encodings are removed.



Evidence concatenates, survivors intersect. Track the set of candidates consistent with everything seen. Appending evidence intersects these sets, in any order, with an empty set absorbing forever. Every answer the machine gives is a read of this one shrinking set, which is why sequential and order-free processing agree and why the internal state has a canonical meaning.³

The least witness settles, even over infinitely many. If candidates are well-ordered and some candidate fits the whole stream, then every smaller candidate dies at some finite time, and past the last such death the least fitting candidate is selected forever. No bound on the number of candidates is used: the ordering does the work of finiteness, which is exactly what survives at the infinite store, and the uniform budgets are exactly what does not.⁴

Each method is one verified statement with its machine instances derived as corollaries; the figures carry worked numbers, so checking the arithmetic in a panel reproduces the corresponding proof's content.

²The count of answer changes along a data stream is classically called mind-change complexity: E. M. Gold, *Language identification in the limit*, 1967; J. Case and C. Smith, *Comparison of identification criteria for machine inductive inference*, 1983.

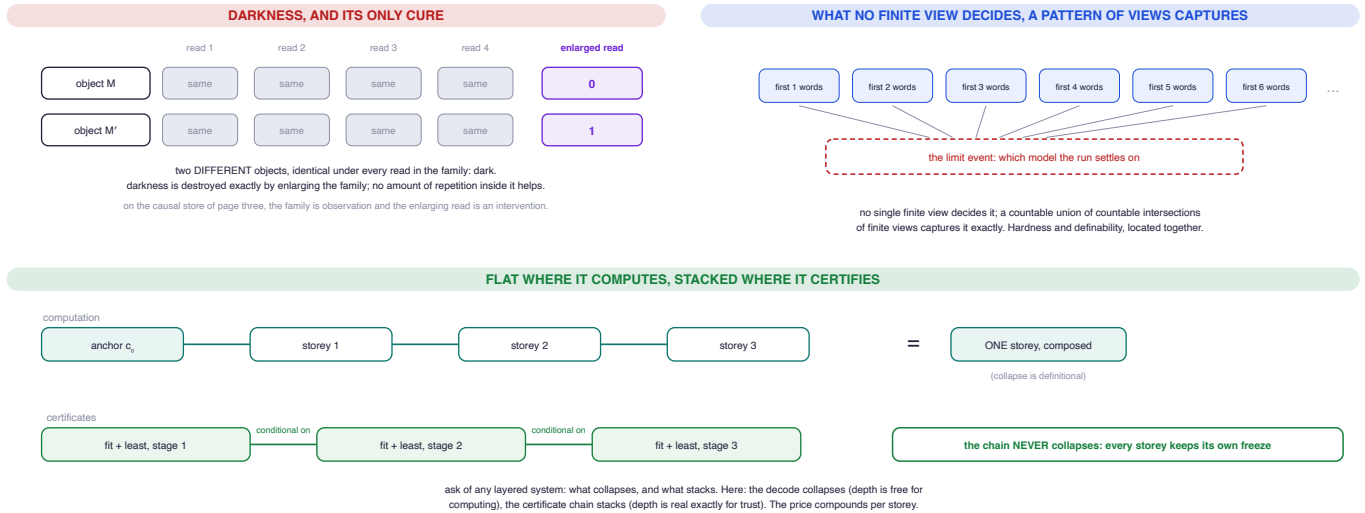
³The consistent set is the version space: T. M. Mitchell, *Generalization as search*, 1982.

⁴Identification by enumeration: E. M. Gold, 1967. The infinite-store positive and its two negative companions are page two's final entry.

Methods, II ways of separating

Darkness, and its only cure. Fix a family of reads. Two different objects with identical values under every read in the family are dark to it, and no repetition of reads inside the family can split them; they are separated exactly when the family is enlarged by a read that tells them apart. Naming the dark pairs locates what a system cannot know now and prices what it must buy to know it. On the causal store the family is observation, the enlarging read is an intervention, and the machine computes the price of the cure in words.⁵

What no finite view decides, a pattern of views captures. Some facts about a process's limit are decided by no finite view, yet are captured exactly by a countable union of countable intersections of finite views. Locating a limit fact at this level is a two-sided result: a hardness floor (no finite view suffices) and a definability ceiling (this pattern suffices), stated together.⁶



Flat where it computes, stacked where it certifies. In a layered system, ask what composition does at each register. Where composing layers collapses definitionally, depth is free and carries no content; whatever refuses to collapse is where depth is real. For the cascade, the decode collapses (any tower of anchorings is one anchoring) while the certificate chain stacks (every storey keeps its own conditional freeze), and the execution price compounds per storey; the dichotomy of page four says depth becomes expressive exactly when modifiers are grammar-bound.⁷

Modify without rebuilding: the one-hole graft. To modify a structured object, take the object's own syntax, puncture it once, and graft the accepted structure into the hole. Because the graft is an operation of the grammar, modified objects inherit every law automatically; nothing about modification needs a new theory. This is the whole content of page four's binding, and it is what restores the compute family to every storey of the cascade.⁸

The frontier. Stated as obligations, in the object's own terms: the sharpness of the graft prices; the scored reading of the gates; the distributional layer over the deterministic mechanisms; and the exact price law of experiments. Each names structure to be built, none weakens a law above.

Everything on these six pages is machine-checked in Lean 4 over Mathlib. Axiom footprint: the standard triple or less, several central laws axiom-free; no unfinished proofs, warnings promoted to errors. The executed instances (pages three, five, six) run inside the checker's decision procedure: the tables and worked panels are evaluated, not typeset claims.

⁵On causal graphs, observational darkness is Markov equivalence: J. Pearl, *Causality*, 2009. The machine's weight stratum (page one) is a second instance: scores are dark to the survivor trajectory.

⁶For the connection between revision counts and this definability level: R. Freivalds and C. Smith, *On the role of procrastination in machine learning*, 1993; M. de Brecht and A. Yamamoto, 2010.

⁷The collapse is Kleisli composition of the retrieval steps: E. Moggi, *Notions of computation and monads*, 1991.

⁸One-hole contexts and their calculus: G. Huet, *The zipper*, 1997; C. McBride, *The derivative of a regular type is its type of one-hole contexts*, 2001.